Frame Partitioning Algorithms

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Masters Thesis Defense

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Outline

1. Problem History and Motivation
   - The Kadison-Singer Problem
   - A Positive Solution to the KS Conjecture
   - What are Finite Frames?
   - Motivation
2. Preliminaries
   - The MSS Theorem
   - Finite Frames
3. Equinorm-Equiangular Frames
   - What are Equinorm-Equiangular Frames?
   - Sums of Tensor Products
   - Norms of Tensor Product Sums
   - Even Split Partition
   - Unequal Partitions
4. Grassmannian Frames
   - What are Grassmannian Frames?
   - Analysis of the Inner and Tensor Products
   - Partitions of the Grassmannian: $m = 2$
   - An Analytic Approach to the $m = 2$ case
   - Partitions of the Grassmannian: $m = 4$
   - Partitions of the Grassmannian: $m = 16$
5. Conclusion
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In 1959 Richard Kadison and Isadore Singer posed the following problem [MSS13, CC06, Wea04]:

Kadison-Singer Problem (KS)

Does every pure state on the (abelian) von Neumann algebra $\mathbb{D}$ of bounded diagonal operators in ℓ_2 have a unique extension to a (pure) state on $B(\ell_2)$, the von Neumann algebra of all bounded linear operators on the Hilbert space ℓ_2?

Originally Kadison thought the answer to this question was negative.

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This research explores ways to partition a special set of vectors known as *finite frames*.

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Finite frames [RJD52] were formally defined by Duffin and Schaeffer in 1952.

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This is relevant because in the fields of Computer Science, Computational Mathematics, and Signal Processing we are always in need of finding algorithms to partition data effectively.

Dot Products and Inner Products

General Inner Product

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- $\langle x, x \rangle$ is real
- $\langle x, y \rangle = \overline{\langle y, x \rangle}$
- $\langle ax, y \rangle = a\langle x, y \rangle$
- $\langle x, y \rangle \overline{\langle x, y \rangle} = |\langle x, y \rangle|^2$

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We will define $\langle \star, \star \rangle = \star \cdot \overline{\star}$, the dot product; therefore

- $\langle x, x \rangle = x \cdot \overline{x} = \|x\|^2 \Rightarrow \|x\|$ is the euclidean norm
- $\langle x, y \rangle = x \cdot \overline{y}$

Eigenvalues and Eigenvectors of a Matrix

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For an $n \times n$ matrix $\mathbf{M}$, scalars λ and nonzero vectors $v \in \mathbb{C}^n$ satisfying $\mathbf{M}v = \lambda v$ are called *eigenvalues* and *eigenvectors* of $\mathbf{M}$.

The eigenvalues are obtained by solving the *characteristic* equation

$$\det(\mathbf{M} - \lambda \mathbf{I}) = 0.$$

Types of Norms

Our research utilizes two types of *norms*:

- The Euclidean vector norm for a vector $x \in \mathbb{C}^n$ defined as $\|x\| = \sqrt{\langle x, x \rangle}$.
- The Induced matrix norm.

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Other norms which go beyond the scope of this research are

- The Matrix 2-norm, $\|\mathbf{M}\|_2 = \sqrt{\lambda_{max}}$, the max eigenvalue of $\mathbf{M}^*\mathbf{M}$
- The Matrix 1-norm, $\|\mathbf{M}\|_1 =$ the largest absolute column sum.
- The Matrix ∞-norm, $\|\mathbf{M}\|_\infty =$ the largest absolute row sum.
- The Frobenius norm, $\|\mathbf{M}\|_F = \sqrt{trace(\mathbf{M}^*\mathbf{M})}$

Matrix Norms

Example

Calculating the Norm of a Hermitian Matrix

Given the real matrix $\mathbf{M} = \begin{bmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{bmatrix}$, then find $\|\mathbf{M}\|$.

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Given the real matrix $\mathbf{M} = \begin{bmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{bmatrix}$, then find $\|\mathbf{M}\|$.

Notice that $\mathbf{M} = \mathbf{M}^T$. We start by finding eigenvalues of $\mathbf{M}$,

$$\det(\mathbf{M} - \lambda \mathbf{I}) = \begin{vmatrix} 1-\lambda & \sqrt{2} \\ \sqrt{2} & 2-\lambda \end{vmatrix}$$

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Therefore, $\|\mathbf{M}\| = 3$. The same procedure applies to complex matrices.

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Induced Matrix Norms

A vector norm that is defined on $\mathbb{C}^p$ for $p = m, n$ *induces* a matrix norm on $\mathbb{C}^{m\times n}$ by setting

$$\|\mathbf{A}\| = \max_{\|x\|=1} \|\mathbf{A}x\|$$

for $\mathbf{A} \in \mathbb{C}^{m\times n}, x \in \mathbb{C}^n$.

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- Note: $\sigma(\mathbf{A})$, the spectrum is the set of all eigenvalues of the matrix $\mathbf{A}$.

Matrix Decomposition

Spectral Decomposition

Given a normal matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$ that has an orthogonal eigenvector basis, it may be decomposed as

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^*$$

where $\mathbf{U}$ is a unitary matrix consisting of eigenvectors of $\mathbf{A}$ and $\mathbf{\Lambda}$ is a diagonal matrix with eigenvalues on the main diagonal.

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- A unitary matrix: $\mathbf{U}\mathbf{U}^* = \mathbf{U}^*\mathbf{U} = \mathbf{I}$

Tensor Products

The Tensor Product

Given two vectors $x, y \in \mathbb{C}^n$ the tensor product (outer product) of two vectors is defined as

$$x \otimes y = xy^* = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} \overline{y}_1 & \overline{y}_2 & \cdots & \overline{y}_n \end{bmatrix}$$

$$= \begin{bmatrix} x_1\overline{y}_1 & x_1\overline{y}_2 & \cdots & x_1\overline{y}_n \\ x_2\overline{y}_1 & x_2\overline{y}_2 & \cdots & x_2\overline{y}_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n\overline{y}_1 & x_n\overline{y}_2 & \cdots & x_n\overline{y}_n \end{bmatrix}$$

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The MSS Theorem

MSS Theorem (*Corollary 1.5* [MSS13])

Let r be a positive integer and $f_1, ..., f_m \in \mathbb{C}^d$ be vectors such that

$$\sum_{i=1}^{m} f_i \otimes f_i = \mathbf{I}$$

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and $\|f_i\|^2 \leqslant \delta$ for all i. Then there exists a partition $\{S_1, ..., S_r\}$ of $[m]$ such that

$$\left\| \sum_{i \in S_j} f_i \otimes f_i \right\| \leqslant \left(\frac{1}{\sqrt{r}} + \sqrt{\delta} \right)^2 \qquad \text{(MSS)}$$

for $j = 1, ..., r$.

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for $j = 1, ..., r$.

Notice: $[m] = \{1, ..., m\} = S_1 \cup S_2 \cup \cdots \cup S_r$ and each $S_j \subset [m]$

A partition $\{S_1, ..., S_r\}$ that satisfies the MSS inequality will be called a *valid* partition.

An Example of the MSS Theorem

Example

Suppose we have the following three vectors (these form the *Mercedes-Benz* frame)

$$\mathcal{F} = \{f_1, f_2, f_3\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

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- We have $\sum_{i=1}^{3} f_i \otimes f_i = \mathbf{I}$.

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- We have $\sum_{i=1}^{3} f_i \otimes f_i = \mathbf{I}$.
- And each $\|f_i\|^2 = \frac{2}{3}$, so $\delta = \frac{2}{3}$.

Therefore, the three given vectors f_1, f_2, f_3 satisfy the hypothesis of the MSS Theorem.

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The MSS Theorem then states there exists a partition for each value of r. Let us find a partition!

- We only have three vectors; hence $[m] = \{1, 2, 3\}$
- Choose the number of partitions $r = 2$
- A simple 2-partition: let $S_1 = \{1\}$ and $S_2 = \{2, 3\}$

$$S_1 \Rightarrow \mathcal{P}_1 = \{f_1\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix} \right\}$$

$$S_2 \Rightarrow \mathcal{P}_2 = \{f_2, f_3\} = \left\{ \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

An Example of the MSS Theorem

Example

- We have $r = 2$ and $\delta = \frac{2}{3}$; therefore the bound described by the MSS inequality is

$$\left(\frac{1}{\sqrt{r}} + \sqrt{\delta}\right)^2 = \left(\frac{1}{\sqrt{2}} + \sqrt{\frac{2}{3}}\right)^2 = \frac{7}{6} + \frac{2}{\sqrt{3}} > 1$$

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- The corresponding norms of the tensor product sums for $\mathcal{P}_1, \mathcal{P}_2$ are

$$\|f_1 \otimes f_1\| = \frac{2}{3}$$
$$\|f_2 \otimes f_2 + f_3 \otimes f_3\| = 1$$

Both of these norms are less than the bound $\frac{7}{6} + \frac{2}{\sqrt{3}}$.
$\Rightarrow \{S_1, S_2\}$ is a *valid* partition of $\mathcal{F}$.

A Summary of the MSS Theorem

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- The theorem states that there *exists* a partition of the vectors.
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Definition: Basis

A linearly independent spanning set for a vector space $\mathcal{V}$ is called a **basis** for $\mathcal{V}$.

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The *standard* basis vectors in $\mathbb{R}^3$ are defined as

$$\mathcal{S} = \{e_1, e_2, e_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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- Linearly independent:

$$c_1 e_1 + c_2 e_2 + c_3 e_3 = 0 \Leftrightarrow \{c_1, c_2, c_3\} = \{0, 0, 0\}$$

Crash Course With Finite Frames

Definition: Finite Frame

A frame for a complex space $\mathbb{C}^n$ is a sequence of vectors $\{f_i\} \subset \mathbb{C}^n$ for which there exist constants $0 < A \leqslant B < \infty$ such that, for every $x \in \mathbb{C}^n$,

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- This definition can be extended to Hilbert spaces $\mathcal{H}$.

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- The vectors in $\mathcal{F}$ still span $\mathbb{R}^3$,
- but the vectors are not linearly independent.
- Notice that the vectors are not all necessarily orthonormal, but we can form frames from orthonormal vector sets.

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To do this we will form a matrix $\mathbf{T}$ where the columns of $\mathbf{T}$ are the frame vectors from $\mathcal{F}$

$$\mathbf{T} = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ f_1 & f_2 & f_3 & f_4 & f_5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The $\mathbf{T}$ matrix is called the *synthesis* operator of the frame.

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Next we find $\mathbf{S} = \mathbf{T}\mathbf{T}^*$, this is called the *frame* operator.

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If we find the eigenvalues of $\mathbf{S}$ we get the following

$$\lambda_1 = 6.19258...$$
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The lower frame bound coincides with the smallest eigenvalue
$\Rightarrow A = 0.807418...$

The upper frame bound conincides with the largest eigenvalue
$\Rightarrow B = 6.19258...$

Why Use Frames?

The MSS inequality can be reformulated in terms of frames.

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Note: $\operatorname{card}(\cdot)$ represents the *cardinality* of a set, the number of elements in a set.

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Therefore, the vectors f_i can be thought of as frame vectors with upper frame bound $B = \operatorname{card}(S_j)\delta$.

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Description of Equinorm-Equiangular Frames

Definition: Equinorm-Equiangular Frames

Let our frame $\mathcal{F} =$

$$\{f_j\}_{j=0}^{m-1} = \sqrt{\frac{2}{m}} \begin{bmatrix} \cos\left(\frac{2\pi j}{m}\right) \\ \sin\left(\frac{2\pi j}{m}\right) \end{bmatrix}_{j=0}^{m-1}$$

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- Each vector has been rotated through by $\frac{2\pi}{m}$ radians

The Mercedes-Benz Frame

Example

Let $m = 3$ and we have

$$\mathcal{F} = \{f_0, f_1, f_2\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

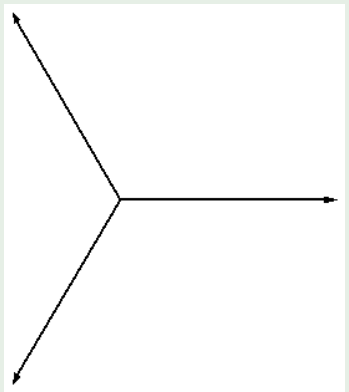


Figure: Mercedes-Benz Frame

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Does this frame satisfy the hypothesis of the MSS Theorem?

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$$= \frac{2}{m} \cdot \frac{1}{2} \sum_{j=0}^{m-1} \begin{bmatrix} 1+\cos\left(\frac{4\pi j}{m}\right) & \sin\left(\frac{4\pi j}{m}\right) \\ \sin\left(\frac{4\pi j}{m}\right) & 1-\cos\left(\frac{4\pi j}{m}\right) \end{bmatrix}$$

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Then the sum of tensor products $\mathbf{F}_{S_1} = \sum_{j \in S_1} f_j \otimes f_j =$

$$\frac{1}{m}\begin{bmatrix} 2 + \cos\left(\frac{4\pi a_1}{m}\right) + \cos\left(\frac{4\pi a_2}{m}\right) & \sin\left(\frac{4\pi a_1}{m}\right) + \sin\left(\frac{4\pi a_2}{m}\right) \\ \sin\left(\frac{4\pi a_1}{m}\right) + \sin\left(\frac{4\pi a_2}{m}\right) & 2 - \cos\left(\frac{4\pi a_1}{m}\right) - \cos\left(\frac{4\pi a_2}{m}\right) \end{bmatrix}$$

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$$\frac{1}{m}\begin{bmatrix} 2 + \cos\left(\frac{4\pi a_1}{m}\right) + \cos\left(\frac{4\pi a_2}{m}\right) & \sin\left(\frac{4\pi a_1}{m}\right) + \sin\left(\frac{4\pi a_2}{m}\right) \\ \sin\left(\frac{4\pi a_1}{m}\right) + \sin\left(\frac{4\pi a_2}{m}\right) & 2 - \cos\left(\frac{4\pi a_1}{m}\right) - \cos\left(\frac{4\pi a_2}{m}\right) \end{bmatrix}$$

We can generalize this for any n number of vectors in a set S_1.

Two-Set Partition Case II: $\mathrm{card}(S_1) = n$

Let $\mathrm{card}(S_1) = n$ then $S_1 = \{a_1, a_2, ..., a_n\}$ for any n number of elements from $[m]$; hence

$$\mathbf{F}_{S_1} = \frac{1}{m} \sum_{j \in S_1} f_j \otimes f_j = \begin{bmatrix} n + \sum_{j \in S_1} \cos\left(\frac{4\pi j}{m}\right) & \sum_{j \in S_1} \sin\left(\frac{4\pi j}{m}\right) \\ \sum_{j \in S_1} \sin\left(\frac{4\pi j}{m}\right) & n - \sum_{j \in S_1} \cos\left(\frac{4\pi j}{m}\right) \end{bmatrix}$$

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Then $S_2 = [m] \setminus S_1$ and $\mathrm{card}(S_2) = m - n = p$ then

$$\mathbf{F}_{S_2} = \frac{1}{m} \sum_{j \in S_2} f_j \otimes f_j = \begin{bmatrix} p + \sum_{j \in S_2} \cos\left(\frac{4\pi j}{m}\right) & \sum_{j \in S_2} \sin\left(\frac{4\pi j}{m}\right) \\ \sum_{j \in S_2} \sin\left(\frac{4\pi j}{m}\right) & p - \sum_{j \in S_2} \cos\left(\frac{4\pi j}{m}\right) \end{bmatrix}$$

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Thus we need to verify

$$\|\mathbf{F}_{S_1}\| \leqslant \left(\frac{1}{\sqrt{2}} + \sqrt{\frac{2}{m}}\right)^2$$

We need to find the norms of $\mathbf{F}_{S_1}$ and $\mathbf{F}_{S_2}$.

Eigenvalues of $\mathbf{F}_{S_1}$

$$\mathbf{F}_{S_1} = \frac{1}{m}\begin{bmatrix} n + \sum_{j\in S_1}\cos\left(\frac{4\pi j}{m}\right) & \sum_{j\in S_1}\sin\left(\frac{4\pi j}{m}\right) \\ \sum_{j\in S_1}\sin\left(\frac{4\pi j}{m}\right) & n - \sum_{j\in S_1}\cos\left(\frac{4\pi j}{m}\right) \end{bmatrix}$$

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To simplify the analysis let

$$\Psi = \sum_{j\in S_1}\cos\left(\frac{4\pi j}{m}\right) \text{ and } \Omega = \sum_{j\in S_1}\sin\left(\frac{4\pi j}{m}\right)$$

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Then

$$\mathbf{F}_{S_1} = \frac{1}{m}\begin{bmatrix} n + \Psi & \Omega \\ \Omega & n - \Psi \end{bmatrix}$$

Eigenvalues of $\mathbf{F}_{S_1}$

$$\det(\mathbf{F}_{S_1} - \lambda \mathbf{I}) = \frac{1}{m} \begin{vmatrix} n + \Psi - m\lambda & \Omega \\ \Omega & n - \Psi - m\lambda \end{vmatrix} = 0$$

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We can repeat the same analysis for S_2 and get

$$\|\mathbf{F}_{S_2}\| = \lambda_{max} = \frac{p}{m} + \frac{\sqrt{\Psi^2 + \Omega^2}}{m}. \tag{2}$$

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The *almost* Even Split Partition: $\mathrm{card}(S_1) = \frac{m+1}{2}$ and $\mathrm{card}(S_2) = \frac{m-1}{2}$.

And like the equal and unequal partitions, we were able to show that the *almost* even split partitions yield valid partitions.

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3. We will also consider a *nice* case where the sets S_1 and S_2 correspond to *consecutive* vectors

$$S_1 = \left\{1, 2, 3, ..., \frac{m}{2}\right\} \Rightarrow \{f_1, f_2, ..., f_{m/2}\}$$

$$S_2 = \left\{\frac{m}{2} + 1, \frac{m}{2} + 2, ..., m\right\} \Rightarrow \{f_{m/2+1}, f_{m/2+2}, ..., f_m\}$$

The Discriminant $\Psi^2 + \Omega^2$

We will examine the S_1 set. Our discriminant becomes

$$\Psi^2 + \Omega^2 = \left(\sum_{j=1}^{m/2} \cos\left(\frac{4\pi j}{m}\right) \right)^2 + \left(\sum_{j=1}^{m/2} \sin\left(\frac{4\pi j}{m}\right) \right)^2$$

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Two summations that will help.

$$\sum_{j=1}^{m/2} \cos(jy) = \sin\left(\frac{my}{4}\right) \cos\left(\frac{(m+2)y}{4}\right) \frac{1}{\sin\left(\frac{y}{2}\right)} \tag{3}$$

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Notice, if we substitute $y = 4\pi/m$ into (3) and (4) then

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Similarly for the set S_2 we get the same result

$$\|\mathbf{F}_{S_1}\| = \|\mathbf{F}_{S_2}\| = \frac{n}{m}$$

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Recall that $n = \frac{m}{2}$; therefore

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Therefore, given an even number of frame vectors, an *even* split of the frame will form two sets that pass the MSS Inequality.

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$\Rightarrow \{S_1, S_2\}$ is a *valid* partition.

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and $n + p = m$ and $n > p$.

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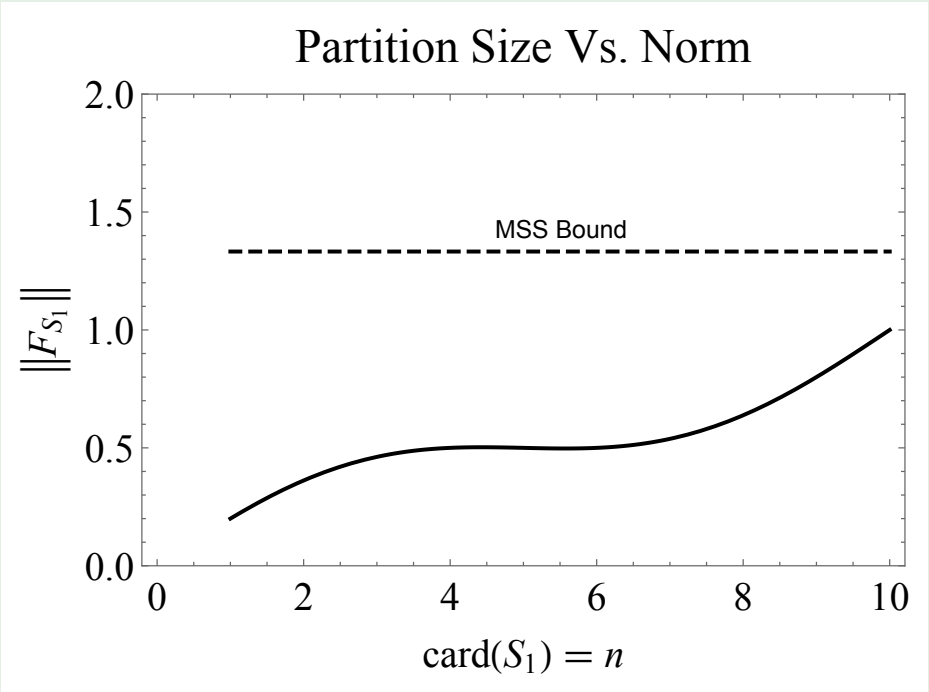
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- Is there any special relationship we can obtain from the inequality above?
- Recall n is the cardinality of S_1 and corresponds to the number of vectors in that partition,
- and m is the total number of vectors in the frame.
- It turns out that for the partition S_1 to be *valid* then n can only take on certain values, and this depends on the size of the frame m.

MSS Theorem Inequality

Example

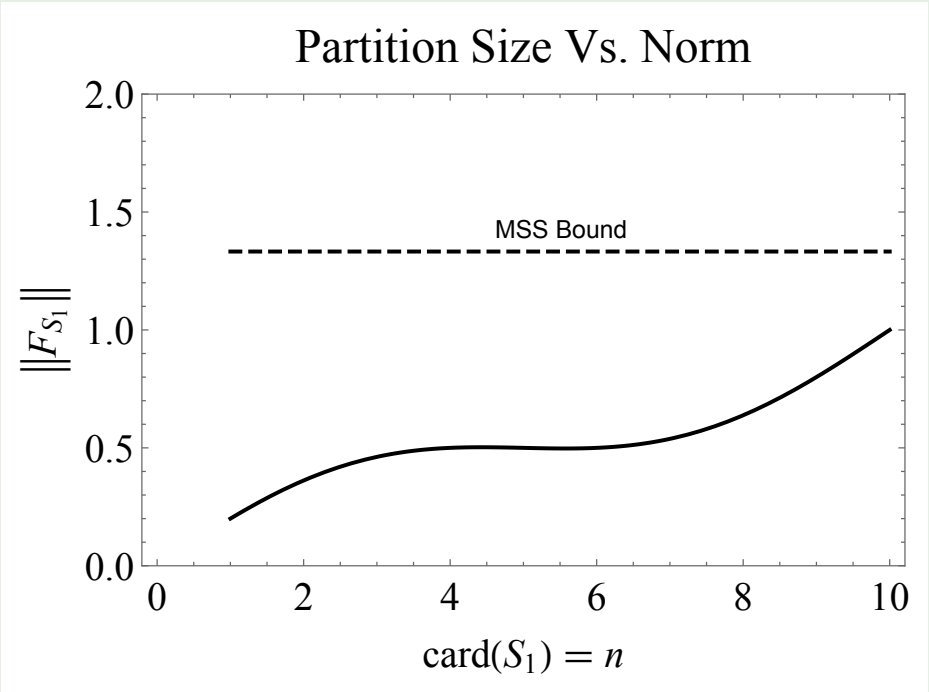
Frame $\mathcal{F}$ has $m = 10$ vectors.



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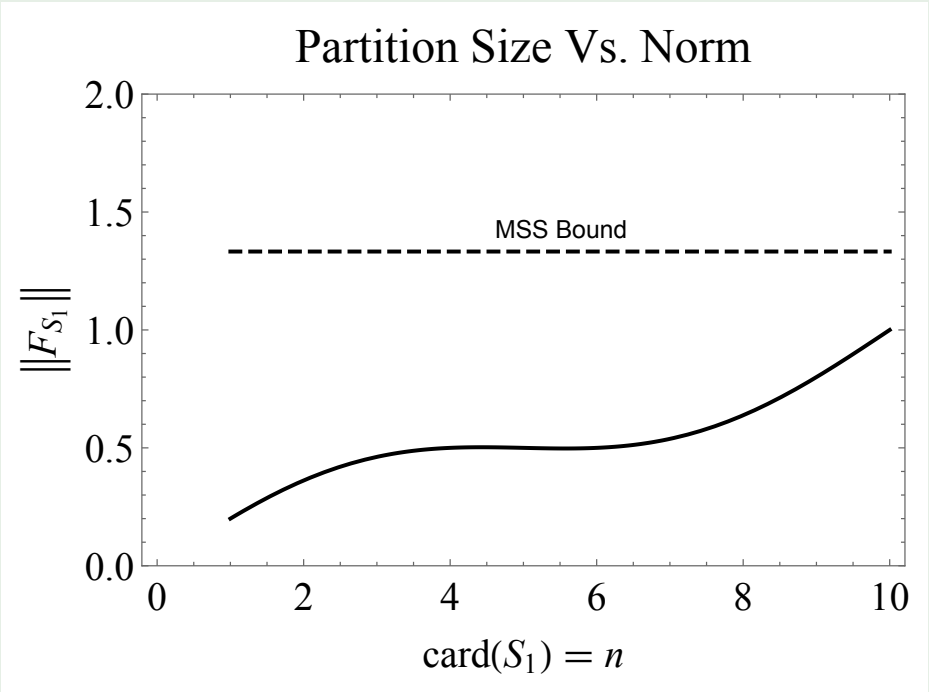


- $\|\mathbf{F}_{S_1}\|$ is always below the bound.

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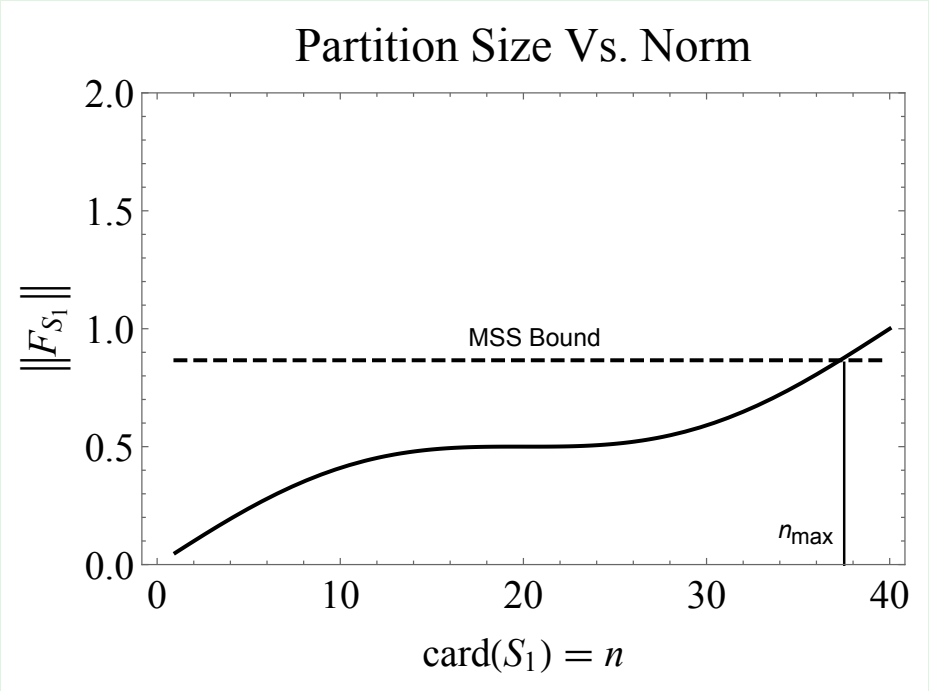


- $\|\mathbf{F}_{S_1}\|$ is always below the bound.
- Any size n of S_1 will be a *valid* partition.

MSS Theorem Inequality

Example

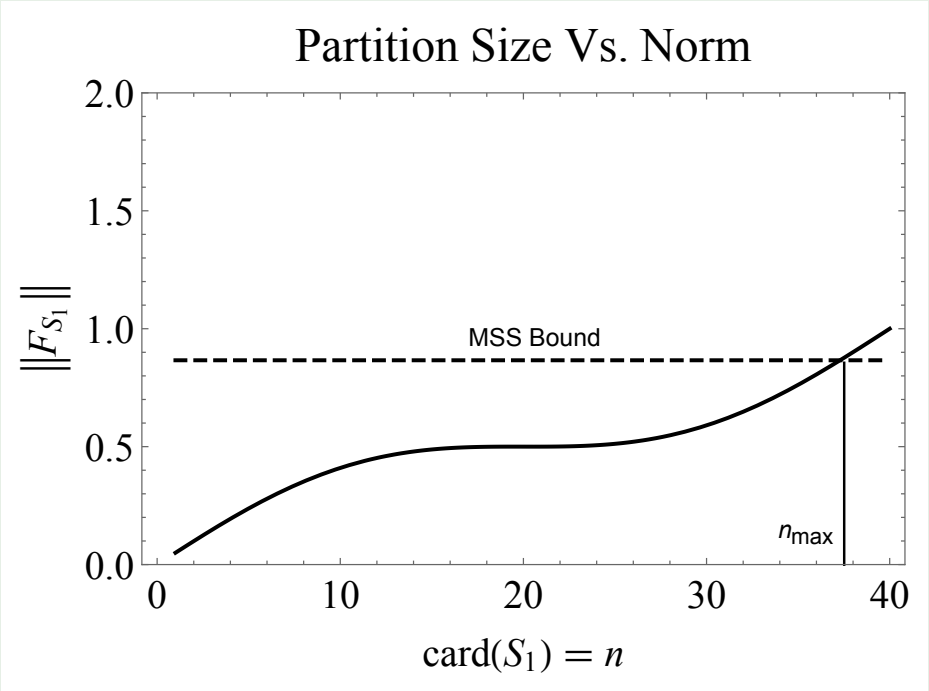
Frame $\mathcal{F}$ has $m = 40$ vectors.



MSS Theorem Inequality

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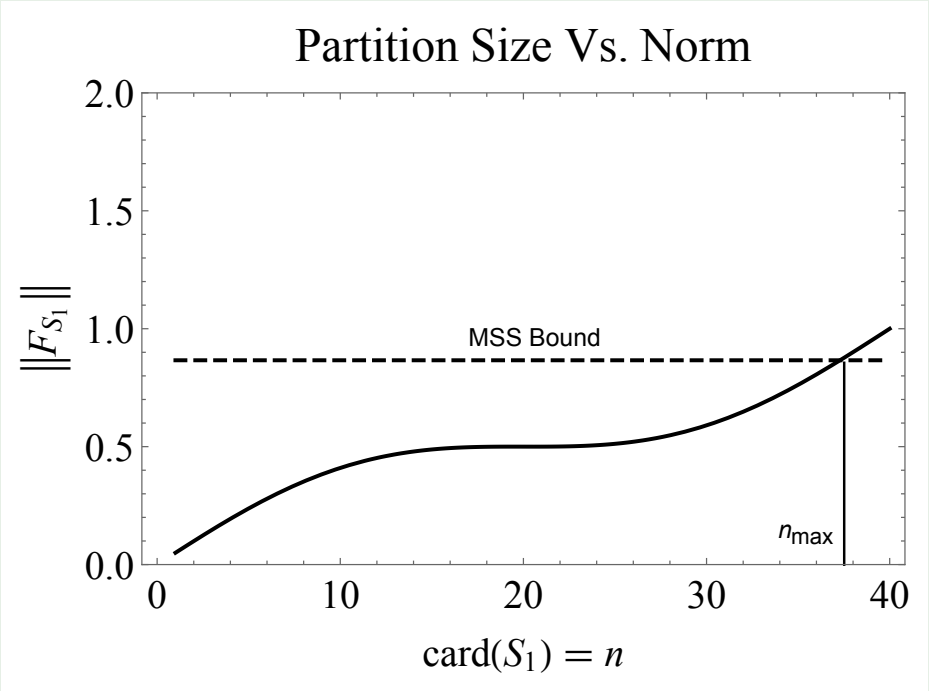


- $\|\mathbf{F}_{S_1}\|$ is not always below the bound.

MSS Theorem Inequality

Example

Frame $\mathcal{F}$ has $m = 40$ vectors.



- $\|\mathbf{F}_{S_1}\|$ is not always below the bound.
- If $\mathrm{card}(S_1) \leqslant n_{max}$ then $\{S_1, S_2\}$ will be a *valid* partition.

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m	n_{max}	$\|\mathbf{F}_{S_1}\|$	MSS Ineq. Bound
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100	84	0.705533...	0.72
200	160	0.64861...	0.651421...

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m	n_{max}	$\|\mathbf{F}_{S_1}\|$	MSS Ineq. Bound
24	23	0.916667...	0.991582...
30	28	0.868123...	0.931815...
40	37	0.852447...	0.866228...
100	84	0.705533...	0.72
200	160	0.64861...	0.651421...
1000	721	0.564478...	0.565245...
10000000	5458762	0.500632654...	0.500632655...

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1. Problem History and Motivation
   - The Kadison-Singer Problem
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Frame Correlation and the Grassmannian

Example

Suppose we have the following set of vectors

$$\mathcal{F} = \{f_j\}_{j=1}^4 = \left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right\}$$

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Next we calculate the magnitude of every possible inner product, with the exception of a vector with itself.

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$$|\langle f_1, f_2 \rangle| = 1$$
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The largest value is called the *maximal frame correlation* $\mathcal{M}(\mathcal{F}) = 1$.

Frame Correlation and the Grassmannian

Definition: Maximal Frame Correlation

For a given *unit norm frame* $\mathcal{F} = \{f_k\}_{k=1}^n$ in $\mathbb{E}^m$ we define the maximal frame correlation $\mathcal{M}(\mathcal{F})$ by

$$\mathcal{M}(\mathcal{F}) = \max_{k,l,k\neq l} \{|\langle f_k, f_l \rangle|\}$$

Frame Correlation and the Grassmannian

Definition: Grassmannian Frame

Given all unit norm frames $\mathcal{F}$ in $\mathbb{E}^m$ we denote

$$\mu = \min\{\mathcal{M}(\mathcal{F})\}.$$

Where μ is the *smallest* of the maximal frame correlations.

A set of vectors $\mathcal{U} = \{u_1, ..., u_n\}$ is a Grassmannian frame if

$$\mathcal{M}(\mathcal{U}) = \mu$$

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- It is known that $\mu \geqslant \sqrt{\frac{n-m}{m(n-1)}}$.
- We will be analyzing *optimal* Grassmannian frames, this occurs when equality holds.

Constructing Grassmannian Frames

- Strohmer and Heath's *optimal* frame construction.

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Form $\mathbf{R}$ Matrix From The Conference Matrix

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Calculate Spectral Decomposition of $\mathbf{R}$

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Extract the Frame Vectors

Constructing Grassmannian Frames

Recursively Form Conference Matrix

Begin with

$$\mathbf{C}_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Let

$$\mathbf{C}_{2m} = \begin{bmatrix} \mathbf{C}_m & \mathbf{C}_m - \mathbf{I}_m \\ \mathbf{C}_m + \mathbf{I}_m & -\mathbf{C}_m \end{bmatrix}$$

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Constructing Grassmannian Frames

Form the **R** Matrix

Define

$$\mathbf{R} := \alpha\mathbf{C} + \mathbf{I}$$

Where,

$$\alpha = \frac{i}{\sqrt{2m-1}}$$

Constructing Grassmannian Frames

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Example

Again let $m = 2$, we have $\mathbf{C}_4$ as before and $\alpha = \frac{i}{\sqrt{3}}$; hence,

$$\mathbf{R} = \begin{bmatrix} 1 & -\frac{i}{\sqrt{3}} & -\frac{i}{\sqrt{3}} & -\frac{i}{\sqrt{3}} \\ \frac{i}{\sqrt{3}} & 1 & \frac{i}{\sqrt{3}} & -\frac{i}{\sqrt{3}} \\ \frac{i}{\sqrt{3}} & -\frac{i}{\sqrt{3}} & 1 & \frac{i}{\sqrt{3}} \\ \frac{i}{\sqrt{3}} & \frac{i}{\sqrt{3}} & -\frac{i}{\sqrt{3}} & 1 \end{bmatrix}$$

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Notice **R** and **C** are Hermitian $\Rightarrow \mathbf{R} = \mathbf{R}^*$

Constructing Grassmannian Frames

Spectral Decomposition of **R** Matrix

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$$\mathbf{R} = \mathbf{W}\Lambda\mathbf{W}^*$$

Constructing Grassmannian Frames

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- **W** is a *unitary* matrix with columns that are the eigen-vectors of the **R** matrix, which are *orthonormal*!

Constructing Grassmannian Frames

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$$\mathbf{R} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^*$$

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Example

If we let $m = 2$ then we have,

$$\mathbf{R} = \mathbf{W} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{W}^*$$

Constructing Grassmannian Frames

Example

In the previous slide we found $\mathbf{\Lambda}$ for $m = 2$, then $\mathbf{W} =$

$$\begin{bmatrix} \frac{1}{6}\left(3i - \sqrt{3}\right) & \frac{i}{\sqrt{6}} & \frac{1}{6}\left(-3i - \sqrt{3}\right) & -\frac{i}{\sqrt{6}} \\ \frac{1}{6}\left(3i + \sqrt{3}\right) & -\frac{i}{\sqrt{6}} & \frac{1}{6}\left(-3i + \sqrt{3}\right) & \frac{i}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{i}{\sqrt{6}} & \frac{1}{\sqrt{3}} & -\frac{i}{\sqrt{6}} \end{bmatrix}$$

Constructing Grassmannian Frames

Example

Isolate the first m columns (left half) of $\mathbf{W}$

$$\left[\begin{array}{cc|cc} \frac{1}{6}\left(3i-\sqrt{3}\right) & \frac{i}{\sqrt{6}} & \frac{1}{6}\left(-3i-\sqrt{3}\right) & -\frac{i}{\sqrt{6}} \\ \frac{1}{6}\left(3i+\sqrt{3}\right) & -\frac{i}{\sqrt{6}} & \frac{1}{6}\left(-3i+\sqrt{3}\right) & \frac{i}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{i}{\sqrt{6}} & \frac{1}{\sqrt{3}} & -\frac{i}{\sqrt{6}} \end{array}\right]$$

Constructing Grassmannian Frames

Example

The rows are the frame vectors,

$$\begin{Bmatrix} \frac{1}{6}\left(3i-\sqrt{3}\right) & \frac{i}{\sqrt{6}} \end{Bmatrix} = f_1^T$$

$$\begin{Bmatrix} \frac{1}{6}\left(3i+\sqrt{3}\right) & -\frac{i}{\sqrt{6}} \end{Bmatrix} = f_2^T$$

$$\begin{Bmatrix} 0 & \frac{1}{\sqrt{2}} \end{Bmatrix} = f_3^T$$

$$\begin{Bmatrix} \frac{1}{\sqrt{3}} & \frac{i}{\sqrt{6}} \end{Bmatrix} = f_4^T$$

Constructing Grassmannian Frames

Extract the Frame Vectors from **W**

Let the first half of each row from **W** define the frame vectors

$$f_k = \{\mathbf{W}_{k,l}\}_{l=1}^{m}$$

Where each $f_k \in \mathbb{C}^m$. And the frame is

$$\mathcal{F} = \{f_k\}_{k=1}^{n}$$

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- There are n rows, these are the frame vectors.
- Hence we have n frame vectors each is in $\mathbb{C}^m$.

Constructing Grassmannian Frames

Recall $n = 2m$, thus

$$\mathbf{W} = \left[\begin{array}{ccc|ccc} \uparrow & & \uparrow & \uparrow & & \uparrow \\ w_1 & \cdots & w_m & w_{m+1} & \cdots & w_n \\ \downarrow & & \downarrow & \downarrow & & \downarrow \end{array}\right]$$

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We will denote the left *half* of the above matrix as

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The rows of $\mathcal{W}$ are the frame vectors; hence

$$f_k = \mathcal{W}^T e_k$$

And $\mathcal{F} = \{f_k\}_{k=1}^n$ with $f_k \in \mathbb{C}^m$.

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Therefore

- $\langle f_k, f_k \rangle = \|f_k\|^2 = \frac{1}{2}$ hence $\|f_k\| = \frac{1}{\sqrt{2}}$
- Since $\langle f_j, f_k \rangle \neq 0$ when $j \neq k$ the frame vectors are not orthogonal!

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$\Rightarrow$ This frame satisfies the hypothesis of the MSS Theorem!

Outline

1. Problem History and Motivation
   - The Kadison-Singer Problem
   - A Positive Solution to the KS Conjecture
   - What are Finite Frames?
   - Motivation
2. Preliminaries
   - The MSS Theorem
   - Finite Frames
3. Equinorm-Equiangular Frames
   - What are Equinorm-Equiangular Frames?
   - Sums of Tensor Products
   - Norms of Tensor Product Sums
   - Even Split Partition
   - Unequal Partitions
4. Grassmannian Frames
   - What are Grassmannian Frames?
   - Analysis of the Inner and Tensor Products
   - Partitions of the Grassmannian: $m = 2$
   - An Analytic Approach to the $m = 2$ case
   - Partitions of the Grassmannian: $m = 4$
   - Partitions of the Grassmannian: $m = 16$
5. Conclusion
6. Credits

The Frame Vectors

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- $\mathrm{card}(\mathcal{F}) = 4$,
- each $f_j \in \mathbb{C}^2$,
- and from the construction

$$\mathcal{F} = \left\{ \begin{bmatrix} \frac{1}{6}(3i - \sqrt{3}) \\ \frac{i}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} \frac{1}{6}(3i + \sqrt{3}) \\ -\frac{i}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{i}{\sqrt{6}} \end{bmatrix} \right\}$$

MSS Theorem Inequality

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4 sets	1.45710...

Norms of Tensor Product Sums Case $m = 4$

- The left column lists every possible subset of $[4] = \{1, 2, 3, 4\}$.

S	$\|\mathbf{F}_S\|$
$\{1\}$	0.5
$\{2\}$	0.5
$\{3\}$	0.5
$\{4\}$	0.5
$\{1, 2\}$	0.788675...
$\{1, 3\}$	0.788675...
$\{1, 4\}$	0.788675...
$\{2, 3\}$	0.788675...
$\{2, 4\}$	0.788675...
$\{3, 4\}$	0.788675...
$\{1, 2, 3\}$	1
$\{1, 2, 4\}$	1
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- Notice we have $2^{2m} - 1 = 15$ possible sets, a power set.

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$\{2, 4\}$	0.788675...
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$\{1, 2, 3\}$	1
$\{1, 2, 4\}$	1
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- The left column lists every possible subset of $[4] = \{1, 2, 3, 4\}$.
- Notice we have $2^{2m} - 1 = 15$ possible sets, a power set.
- We can calculate every possible norm of tensor product sums, this is the right column.

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$\{1\}$	0.5
$\{2\}$	0.5
$\{3\}$	0.5
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Norms of Tensor Product Sums Case $m = 4$

Example

Suppose we fix $r = 2$.

Then $[4] = S_1 \cup S_2$

Our partition is $\{S_1, S_2\}$.

$$S_1 = \{1, 2\}$$
$$S_2 = \{3, 4\}$$

The corresponding norms are

$$\|\mathbf{F}_{S_1}\| = 0.788675 < 2$$
$$\|\mathbf{F}_{S_2}\| = 0.788675 < 2$$

$\Rightarrow$ This is a *valid* partition.

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Norms of Tensor Product Sums Case $m = 4$

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A different partition

$S_1 = \{1\}$

$S_2 = \{2, 3, 4\}$

The corresponding norm

$$\|\mathbf{F}_{S_1}\| = 0.5 < 2$$
$$\|\mathbf{F}_{S_2}\| = 1 < 2$$

$\Rightarrow$ This is a *valid* partition.

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Norms of Tensor Product Sums Case $m = 4$

Example

Suppose we fix $r = 3$.

Then $[4] = S_1 \cup S_2 \cup S_3$

$S_1 = \{1\}$

$S_2 = \{2\}$

$S_3 = \{3, 4\}$

The corresponding norms

$$\|\mathbf{F}_{S_1}\| = 0.5 < 1.64982...$$
$$\|\mathbf{F}_{S_2}\| = 0.5 < 1.64982...$$
$$\|\mathbf{F}_{S_3}\| = 0.788675... < 1.64982...$$

$\Rightarrow$ This is a *valid* partition.

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Norms of Tensor Product Sums Case $m = 4$

Example

Suppose we fix $r = 4$.

Then $[4] = S_1 \cup S_2 \cup S_3 \cup S_4$

$S_1 = \{1\}$

$S_2 = \{2\}$

$S_3 = \{3\}$

$S_4 = \{4\}$

The corresponding norms are all equal!

$$\|\mathbf{F}_{S_1}\| = \cdots = \|\mathbf{F}_{S_4}\| = 0.5 < 1.45710...$$

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MSS Theorem Inequality

We exhaustively compared all possible norms to the MSS bounds and arrived at the following

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Theorem

Given an optimal Grassmannian frame $\mathcal{F} = \{f_1, f_2, f_3, f_4\}$, any partition of $\mathcal{F}$ will be a valid partition to the MSS theorem.

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A Review of Properties

- Our frame is $\mathcal{F} = \{f_1, f_2, f_3, f_4\}$.
- Each vector f_k is equal to a row from $\mathcal{W}$.
- For each vector $\|f_k\|^2 = \frac{1}{2}$.
- The inner product between two vectors, for $m = 2$ is

$$\langle f_k, f_j \rangle = \pm \frac{i}{2\sqrt{3}}$$

Analysis

Suppose we take f_1 and multiply it by a unitary matrix $\mathbf{U}$. We define the new vector as

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This *rotates* the vector f_1 to a standard position and maintains the norm.

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Multiply f_2 by $\mathbf{U}$ to get $\tilde{f}_2 = \{b, c\}^T$ and

$$\|\tilde{f}_2\|^2 = b\overline{b} + c\overline{c} = \frac{1}{2}$$

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Notice $\langle \tilde{f_1}, \tilde{f_2} \rangle = \{a, 0\} \cdot \{\overline{b}, \overline{c}\} = a\overline{b} = \langle f_1, f_2 \rangle$, where

$$\overline{b} = \frac{\langle f_1, f_2 \rangle}{a} \text{ and } b = \frac{\langle f_2, f_1 \rangle}{\overline{a}}.$$

Analysis

We combine our results to find $b\overline{b}$ and $c\overline{c}$ to get

$$b\overline{b} = \frac{|\langle f_1, f_2 \rangle|^2}{a\overline{a}} = \frac{1}{6}$$

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Let us calculate the norm of the following matrix

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Since we multiplied each vector by a unitary matrix our matrix $\mathbf{F}$ is preserved. Hence

$$\mathbf{F} = \tilde{f}_1 \otimes \tilde{f}_1 + \tilde{f}_2 \otimes \tilde{f}_2 = \begin{bmatrix} a\overline{a} + b\overline{b} & b\overline{c} \\ \overline{b}c & c\overline{c} \end{bmatrix}.$$

Solve the characteristic equation

$$(a\overline{a} + b\overline{b} - \lambda)(c\overline{c} - \lambda) - b\overline{b}c\overline{c} = 0.$$

Analysis

We know the quantities $a\bar{a}$, $b\bar{b}$, and $c\bar{c}$ and our equation becomes

$$\lambda^2 - \lambda + \frac{1}{6} = 0$$

The solutions to this equation are $\{\lambda_1, \lambda_2\} = \{\frac{1}{6}\left(3 - \sqrt{3}\right), \frac{1}{6}\left(3 + \sqrt{3}\right)\}$.

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$$\|\mathbf{F}\| = \max\{\lambda_1, \lambda_2\} = \frac{1}{6}\left(3 + \sqrt{3}\right) = 0.788675\ldots$$

This is exactly the value we found numerically for two vectors in a subset.

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This is exactly the value we found numerically for two vectors in a subset. We can follow the same pattern of analysis to find $\tilde{f}_3 = \{d, e\}^T$, where

$$d\overline{d} = \frac{1}{6} \text{ and } e\overline{e} = \frac{1}{3}.$$

Analysis

If we calculate the matrix $\mathbf{F}$ for three vectors we get

$$\mathbf{F} = f_1 \otimes f_1 + f_2 \otimes f_2 + f_3 \otimes f_3.$$

Again since we multiplied each vector by a unitary matrix our matrix $\mathbf{F}$ is preserved; therefore,

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$$\lambda^2 - \frac{3}{2}\lambda + \frac{1}{2} = 0.$$

The solutions to this equation are $\{\lambda_1, \lambda_2\} = \{\frac{1}{2}, 1\}$.
Therefore

$$\|\mathbf{F}\| = \max\{\lambda_1, \lambda_2\} = 1.$$

This verifies our numerical results for three vectors in a subset.

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- we need to find partitions of the 8 vector set $\{f_1, f_2, f_3, f_4, f_5, f_6, f_7, f_8\}$,

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Unlike the previous case, the number of possible sets that may be formed from $\{1, ..., 8\}$ is $2^{2m} - 1 = 255$.

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Below is a table of norms corresponding to different cardinalities of subsets $S \subset \{1, ..., 8\}$.

$\mathrm{card}(S)$	$\|\mathbf{F}_S\|$
1	0.5
2	0.688982...
3	0.827327...
4	0.827327... or 0.956243
$\geqslant 5$	1

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- For certain values of $\mathrm{card}(S)$ the corresponding norm takes on particular values,

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2	0.688982...
3	0.827327...
4	0.827327... or 0.956243
$\geqslant 5$	1

- For certain values of $\mathrm{card}(S)$ the corresponding norm takes on particular values,
- If $\mathrm{card}(S) \geqslant 5$ the corresponding norm is always 1, interestingly 5 is one more than half of m.

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All the norms fall well below the above MSS bound $\frac{9}{8}$.
$\Rightarrow$ Numerically confirmed that any partition of the Grassmannian frame for $m = 4$ is a valid partition.

Outline

1. Problem History and Motivation
   - The Kadison-Singer Problem
   - A Positive Solution to the KS Conjecture
   - What are Finite Frames?
   - Motivation
2. Preliminaries
   - The MSS Theorem
   - Finite Frames
3. Equinorm-Equiangular Frames
   - What are Equinorm-Equiangular Frames?
   - Sums of Tensor Products
   - Norms of Tensor Product Sums
   - Even Split Partition
   - Unequal Partitions
4. Grassmannian Frames
   - What are Grassmannian Frames?
   - Analysis of the Inner and Tensor Products
   - Partitions of the Grassmannian: $m = 2$
   - An Analytic Approach to the $m = 2$ case
   - Partitions of the Grassmannian: $m = 4$
   - Partitions of the Grassmannian: $m = 16$
5. Conclusion
6. Credits

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The number of possible sets that may be formed from $\{1, ..., 32\}$ is $2^{2m} - 1 = 4294967295$

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For this particular case we actually find partitions that are *not valid*.

MSS Theorem Inequality

Example

Let $r = 17$, this means we partition $[32]$ into 17 subsets; hence

$$[32] = S_1 \cup S_2 \cup \cdots \cup S_{17}$$

Use a *consecutive* vector partitioning.

The MSS bound is

$$\left(\frac{1}{\sqrt{17}} + \frac{1}{\sqrt{2}}\right)^2 = 0.901821...$$

All the norms are below this bound. $\Rightarrow$ This is a valid partition.

$S_k = \{...\}$	$\|\mathbf{F}_{S_k}\|$
$S_1 = \{1, ..., 16\}$	0.847804...
$S_2 = \{17\}$	0.5
$S_3 = \{18\}$	0.5
$S_4 = \{19\}$	0.5
$S_5 = \{20\}$	0.5
$S_6 = \{21\}$	0.5
$S_7 = \{22\}$	0.5
$S_8 = \{23\}$	0.5
$S_9 = \{24\}$	0.5
$S_{10} = \{25\}$	0.5
$S_{11} = \{26\}$	0.5
$S_{12} = \{27\}$	0.5
$S_{13} = \{28\}$	0.5
$S_{14} = \{29\}$	0.5
$S_{15} = \{30\}$	0.5
$S_{16} = \{31\}$	0.5
$S_{17} = \{32\}$	0.5

MSS Theorem Inequality

Example

Now examine a *non-consecutive* vector partitioning.

Let
$S_1 = \{2, 5, 7, 8, 9, 10, 11, 13, 14, 16, 17, 19, 21, 25, 27, 31\}$.

The MSS bound is still

$$\left(\frac{1}{\sqrt{17}} + \frac{1}{\sqrt{2}}\right)^2 = 0.901821...$$

One norm is not below this bound.

$\Rightarrow$ This is NOT a valid partition.

$S_k = \{...\}$	$\|\mathbf{F}_{S_k}\|$
$S_1 = \{...\}$	0.997832...
$S_2 = \{12\}$	0.5
$S_3 = \{22\}$	0.5
$S_4 = \{4\}$	0.5
$S_5 = \{23\}$	0.5
$S_6 = \{32\}$	0.5
$S_7 = \{3\}$	0.5
$S_8 = \{1\}$	0.5
$S_9 = \{18\}$	0.5
$S_{10} = \{30\}$	0.5
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$S_{14} = \{26\}$	0.5
$S_{15} = \{28\}$	0.5
$S_{16} = \{24\}$	0.5
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Summary of Grassmannian Partitions

- For Grassmannian frames formed for $m = 2,4$, and 8, any partitioning will be valid.
- For the Grassmannian formed for $m = 16$...
  - Consecutive vector partitions appear to yield valid partitions.
  - Non-consecutive vector partitions appear to yield non valid partitions.

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- One of the difficulties with Grassmannian frames is obtaining formulas for the eigenvalues of partial tensor product sums $\|\mathbf{F}_S\|$.
- Obtaining these formulas will allows us to analyze partitions for Grassmannian frames when $m \geqslant 16$.
- Ultimately, this will help us find an *efficient* algorithm to partitioning the frames.

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